Empirically Estimated Uncertainty-Guided Iterative Decoding Strategy of Diffusion Models for Time Series Forecasting

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Overview

We propose a novel **decoding strategy** for **time series diffusion models** by introducing **uncertainty sampling**, which iteratively resamples points with high uncertainty, improving forecasts based on uncertainty estimates.

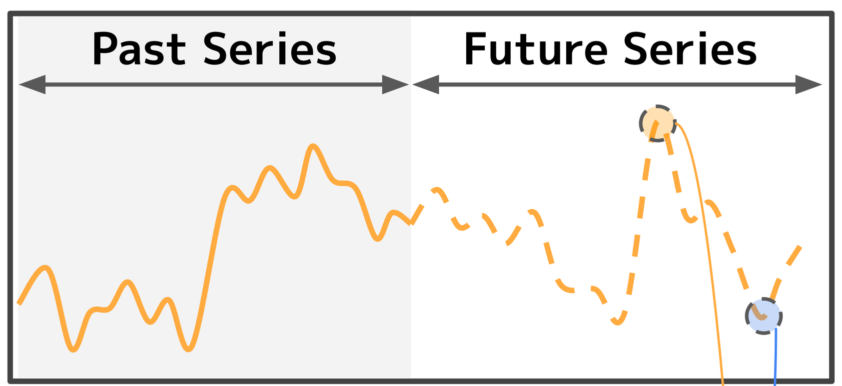
Results: Performance improvements on 7 out of 8 benchmark datasets.
- No improvement was seen in a noisy dataset.

Future Work: Apply to other diffusion models.

Time Series Forecasting (TSF)

Forecast future series from past series.
- Deep learning shows high performance in TSF [1].
- Diffusion model [2] excels in Time Series Completion [3].

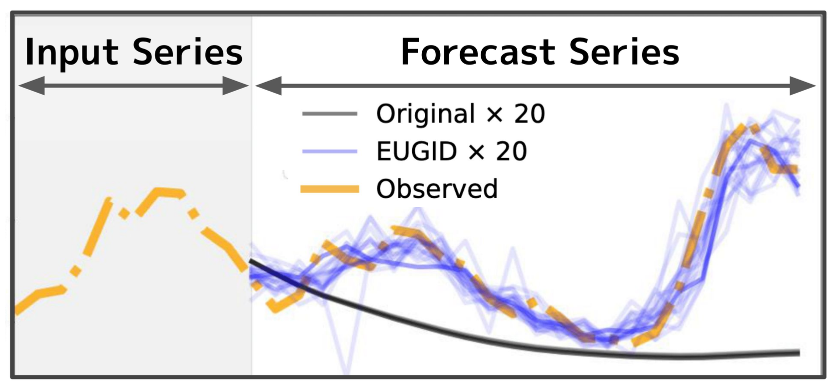
⇔ TSF with diffusion models still lacks high performance [4].
⇒ Analyzing failure cases in TSF to unlock diffusion model potential.



Preliminary Experiment

- Diverse generation, but **deviation from true series**.
- Possibly sampling data with extremely high probability in the learned distribution.

Decoding likely causes deviated outputs.
⇒ Need diverse, accurate predictions.

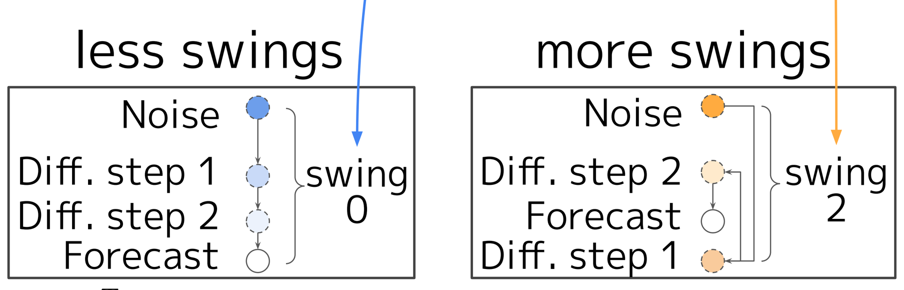


Proposed Method

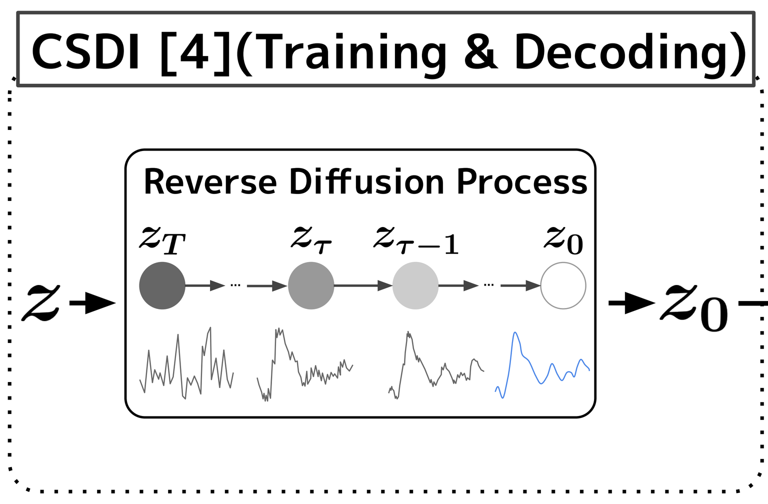
EUGID;
Empirically estimated **U**ncertainty-**G**uided **I**terative **D**ecoding strategy

EUGID Process
- The **Forward Process** adds noise (**ε**) based on the estimated uncertainty to maintain prediction diversity [5].
- The **Reverse Process** removes noise to sample predictions while iteratively refining low-confidence points, leading to improved performance [6].

Estimating Empirical Uncertainty Distribution
Estimate uncertainty as variance $\hat{\alpha}_l$ of predictions at each time step l.
- **Linear/Exponential** estimators assume that points in distant future has larger uncertainty.
- **History-based** estimator assumes that points showing larger swings during reverse diffusion indicate higher uncertainty.

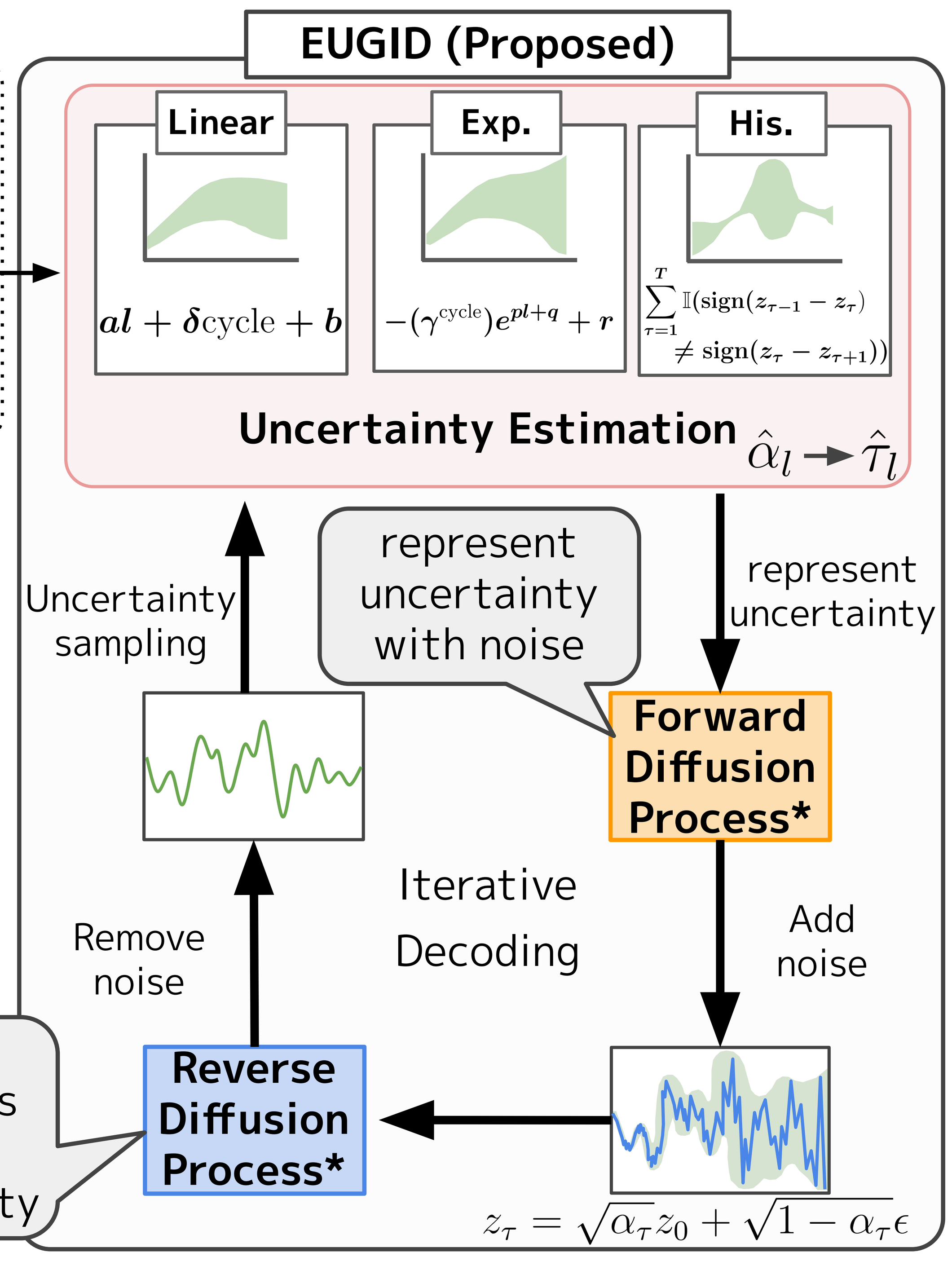


$$\sum_{\tau=1}^{T} \mathbb{I}(\mathrm{sign}(z_{\tau-1} - z_\tau) \neq \mathrm{sign}(z_\tau - z_{\tau+1}))$$

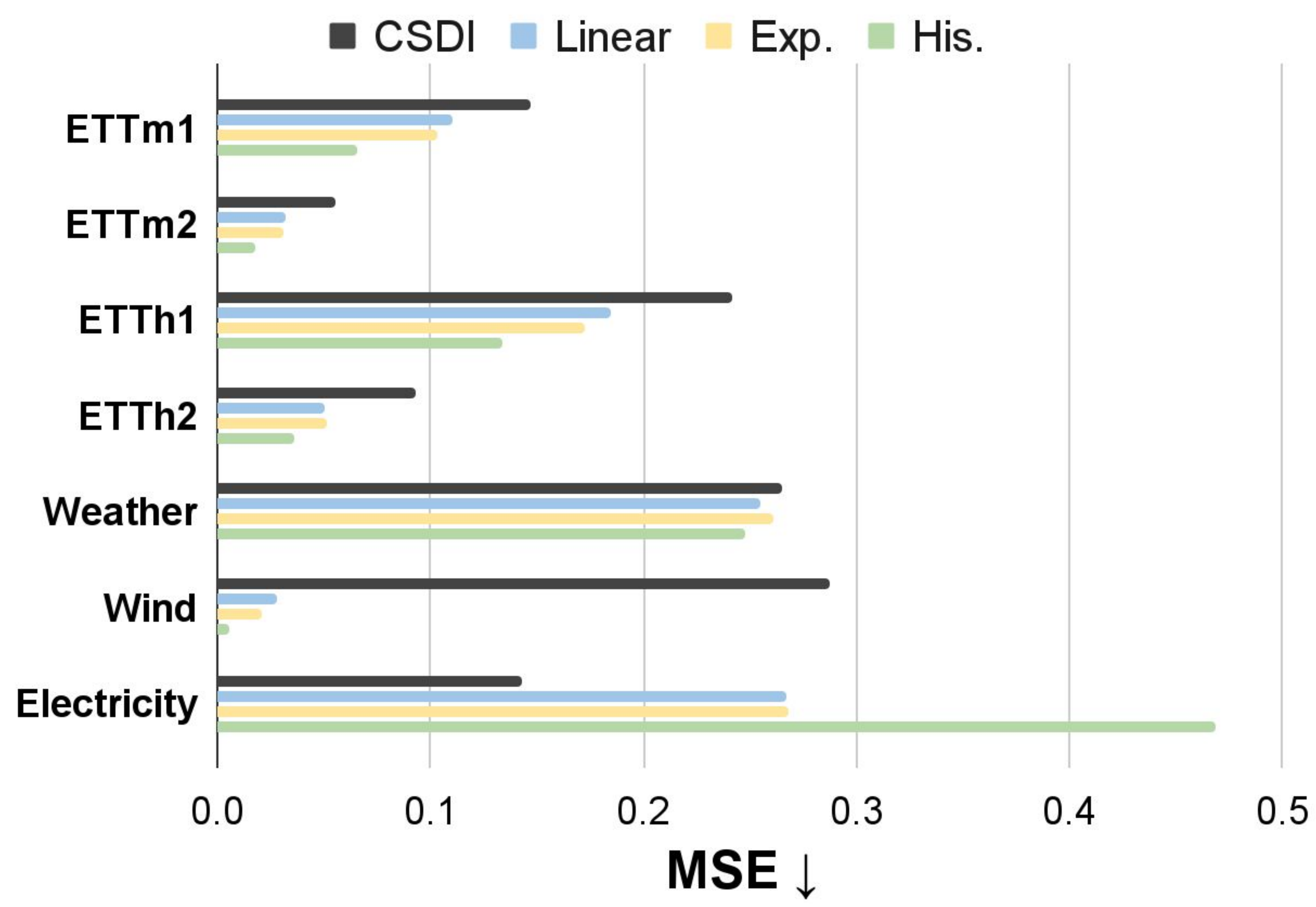


1. Estimate uncertainty as prediction variance $\hat{\alpha}_l$ at each time step l
2. Add noise in forward diffusion process based on estimated uncertainty $\hat{\alpha}_l$
3. Remove noise in reverse diffusion process
4. Repeat steps 1–3 iteratively about 40 times

params
- a: slope
- δ: scaling
- b: bias
- γ: decay
- p: power
- q: shift
- r: offset



Main Results

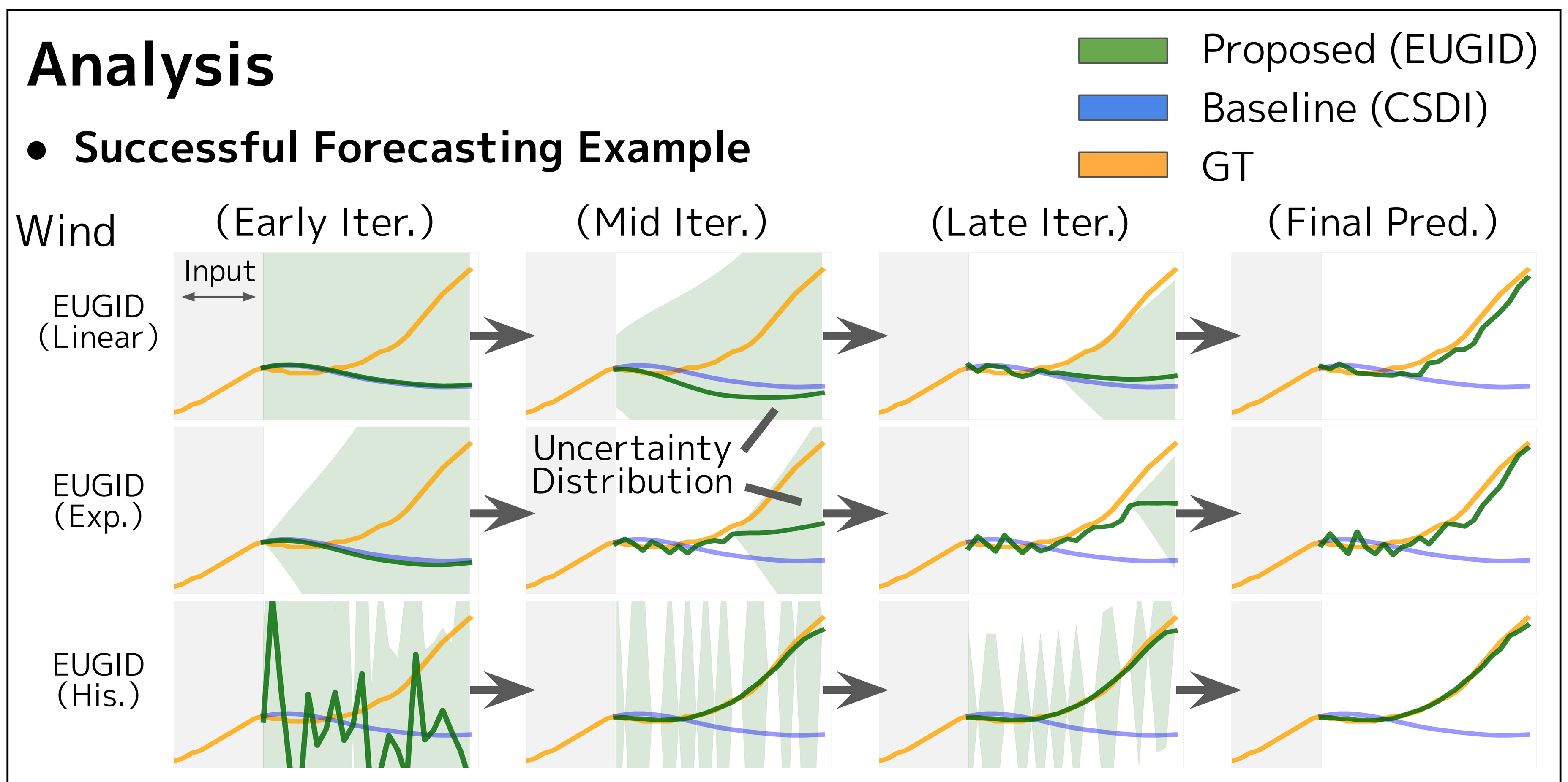


Experimental Setup
- 8 standard TSF datasets [4, 7]
  - Evaluation metric: MSE
  - Input length: 168
  - Forecast horizon: 24
- Baseline: CSDI (Conditional Score-based Diffusion Model) [4]

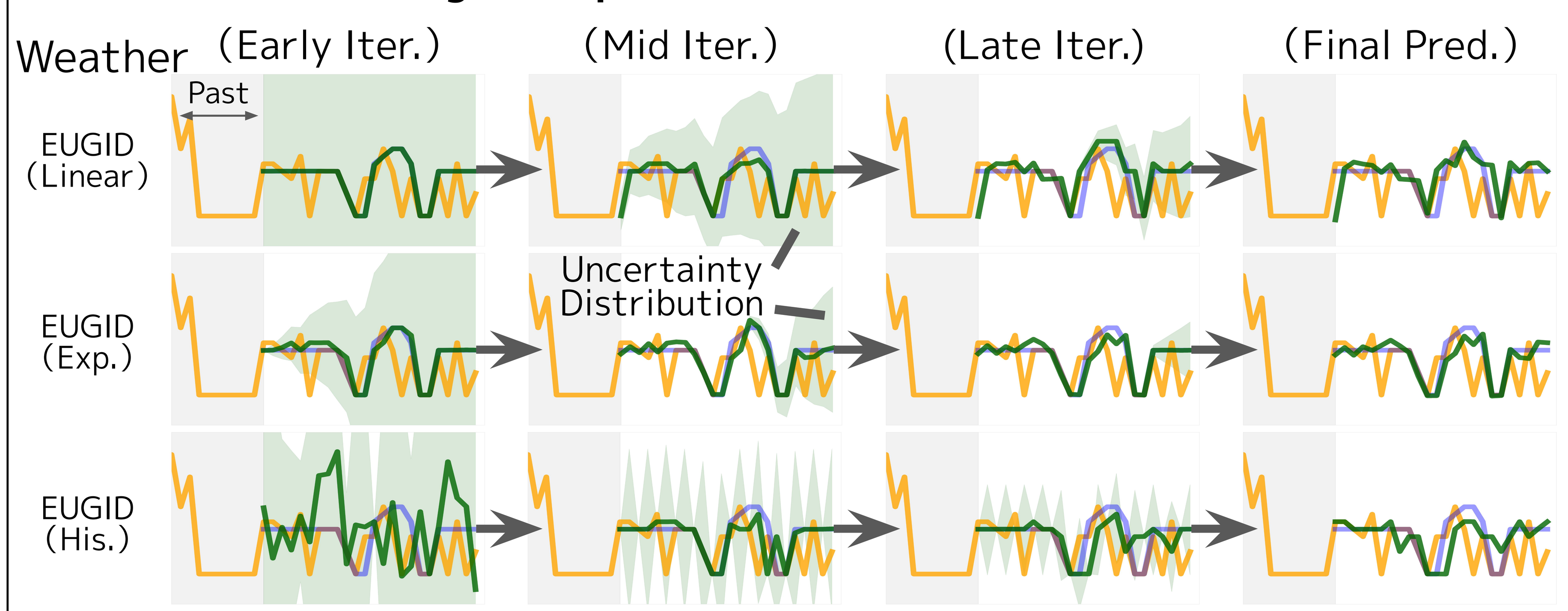
Results
- EUGID significantly improved CSDI on 7/8 datasets. (except Electricity)
- ETTm2: ~67% lower MSE.
- Uncertainty Estimation; Linear ≈ Exponential, while **History-based** estimator works best with EUGID.

Analysis

- **Successful Forecasting Example**



- **Failed Forecasting Example**



- **Analysis of Failure**
  - Electricity: High kurtosis (noisy) → limited performance

Datasets	Electricity	ETTm1	ETTm2	ETTh1	ETTh2	Weather	Wind
Kurtosis	**3.987**	0.413	-0.746	-1.591	0.257	-1.118	-0.340

References

[1] Liu et al., iTransformer: Inverted Transformers Are Effective for Time Series Forecasting, ICLR 2024.
[2] Ho et al., Denoising Diffusion Probabilistic Models, NIPS 2020.
[3] Kollovieh et al., Predict, Refine, Synthesize: Self-Guiding Diffusion Models for Probabilistic Time Series Forecasting, NIPS 2023.
[4] Tashiro et al., CSDI: conditional score-based diffusion models for probabilistic time series imputation, NIPS 2021.
[5] Sadat et al., CADS: Unleashing the Diversity of Diffusion Models through Condition-Annealed Sampling, ICLR 2024.
[6] Raj et al., Uncertainty Sampling for Active Learning, PMLR 2023.
[7] Shen et al., Non-autoregressive conditional diffusion models for time series prediction, ICML 2023.